

Profile

Machine learning engineer with a strong base in fine-tuning, VLMs, agentic systems, and applying ML in specialist domains. Outside my day-to-day work, I compete on Kaggle to explore unfamiliar problems, learn quickly from real constraints, and bring those lessons back to work when building models and tools.

Education

- University of Cambridge, MPhil in Data Intensive Science (Distinction)2024/10 - 2025/07
Advanced study in statistics, machine learning, high-performance computing, and data-intensive scientific applications.
- University of Edinburgh, BA in Mathematics (First Class Honours)2020/09 - 2024/05
Mathematics degree with emphasis on statistics, machine learning, and mathematical foundations for modelling.

Work Experience

- Viridien | Machine Learning Engineer2025/08 – Present
 - VLM document classification:** Fine-tuned vision-language models for oil & gas technical documents using LoRA SFT, RLVR methods (GRPO/GSPO), and prompt optimization with ACE-Agent and DSPy. Improved F1 / accuracy from 0.47 / 0.74 to 0.78 / 0.85 at comparable model size.
 - Agentic label QC:** Built a QC system for noisy technical-document labels with domain experts. Converted expert review patterns into vision-enabled checks, domain “skills.md” knowledge, and a critic-validator loop to make label decisions easier to audit.
 - Agent terminal-use training:** Fine-tuned Qwen models on multi-GPU infrastructure with Harbor, SGLang, and AREAL to improve terminal-use behavior, in collaboration with CAMEL-AI.
- Autodesk | Intern, Technology Consultant (AI/ML)2024/06 – 2024/09
 - VLM benchmarking for CAM:** Created a benchmark dataset for evaluating how vision-language models understand Computer-Aided Manufacturing workflows. The benchmark supported model selection for Fusion agents and gives the team a repeatable way to test new model releases.
- Business-Intelligence of Oriental Nations Corporation Ltd | Intern, Algorithm Engineer2023/06 – 2023/09
 - Manufacturing ML:** Trained machine learning models for manufacturing applications using scikit-learn and XGBoost.

Selected Competitions & Hackathons

- Kaggle Hackathon Honorable Mention (8/321) - [Solution Write-up]May 2026
Fine-tuned Gemma 2 2B Instruct for reasoning with a curated data mixture, LoRA reasoning SFT, and DPO based on the Delta Learning Hypothesis. Trained on Google TPU.
- Kaggle Competition Master (Ranked 842 worldwide; 1x Gold, 2x Silver).Feb 2026
- Kaggle Competition Gold Medal (13/3357) - [Solution Write-up]Feb 2026
Built a Rust solver for irregular 2D bin packing, placing non-convex shapes into the smallest possible bounding square across hundreds of configurations. Combined simulated annealing, local search, and constraint relaxation over the 3-month competition.
- Kaggle Competition Silver Medal (19/1151) - Presented at NeurIPS 2025 Special Session [Solution Presentation]Oct 2025
Built a program-synthesis agent with an evolutionary database to solve ARC-AGI tasks while optimizing for minimal code length.
- Mistral AI Award - Hack the Law HackathonJun 2025
Built a Mixtral-based multi-agent system to extract, classify, and link regulated activities and entities from legal documents.
- 1st Place - Infosys Cambridge AI Centre HackathonMay 2025
Built **Memory Plus**, an MCP memory layer for LLMs with a local vector store and RAG. (50+ GitHub stars; 8k downloads)

Publications

- SeLoRA: Self-Expanding LoRA for High-Quality and Efficient Medical Image SynthesisFeb 2026
Expert Systems with Applications • Project Page
- Semi-Supervised Medical Image Segmentation via Knowledge Mining from Large ModelsJan 2026
arXiv Preprint • Accepted to IEEE International Symposium on Biomedical Imaging (ISBI)

Skills

AI/ML	Fine-tuning, LoRA, RLVR/GRPO, DPO, prompt optimization, agent evaluation, RAG, synthetic data.
Languages & Tools	Python, Git, Linux, Docker, Neo4j, Qdrant, SQL, LaTeX.
Libraries	vLLM, FastAPI, PyTorch, scikit-learn, pandas, SciPy, Plotly, XGBoost, LightGBM.
Math	Statistics, probability, stochastic processes, time series analysis, probabilistic ML, Bayesian analysis.